Terrain, crop suitability and climate

Controls: Ruggedness and Elevation; Wheat and Maize Suitability; Temperature and Precipitation.

1. What a selected box adds. These controls account for observed differences in physical geography. Each box adds several separate variables, collected in the vector X. The regression estimates a separate coefficient for each variable; selecting all three boxes adds seven variables. The controls do not form one geography score. In ordinary County or MSA regressions they enter directly. In county-adjusted CH, their effects are estimated from counties and enter the regional index through the county contributions.

2. Terrain and crop suitability. Ruggedness and Elevation adds the Nunn–Puga terrain ruggedness measure and PRISM mean elevation. Ruggedness describes how uneven the terrain is; elevation measures height above sea level. Both are measured in metres.

Wheat and Maize Suitability adds two GAEZ indices. They describe rainfed production with low inputs under 1981–2010 conditions. Higher values indicate land more suitable for growing that crop. These are measures of growing conditions, not observed production or the share of land actually farmed. Source values range from zero to 10,000. The regression divides each index by 1,000. A one-unit increase in the regression variable therefore means a 1,000-point increase in the source index.

3. Climate and geographic averaging. Temperature and Precipitation adds January mean temperature, July mean temperature, and annual precipitation. Temperature is in degrees Celsius and precipitation in millimetres. The PRISM climate measures describe the 1991–2020 average. They remain fixed when the modern outcome year or historical census year changes.

Each terrain, crop and climate variable is averaged over the county or MSA using geographic area weights. A larger piece of land receives more weight than a smaller piece. Population and employment do not determine these weights. For MSAs, the averages are computed over the MSA footprint, not by taking an unweighted average of county values.

4. Coverage and interpretation. These measures cover the contiguous United States. Area averages use the grid cells with usable values; a unit with no usable coverage remains missing. Selecting a variable with missing values can remove observations. In county-adjusted CH, a required missing input in a positive-employment county removes the whole MSA or state. The reported density coefficient is conditional on the selected controls. Adding controls can change both the comparison being made and the sample, so compare observation counts as well as coefficients.